

Bin Location **Serial Numbers and Batches**

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the course: Managing Serial Numbers and Batches in Bin Locations.

This course is a part of a series of courses available for the Bin Locations topic.

In this course we will cover the processes of purchasing, selling and managing serial number or batch managed items in a bin location managed warehouse.

Objectives



At the end of this topic, you will be able to:

- Describe how to allocate bin locations for serial numbers and batches in the sales and purchasing processes.

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Agenda

- Business example
- Purchasing process
- Sales process
- Differences in Batches

This is the agenda for this course.
Let us start with a business example.

Business Scenario

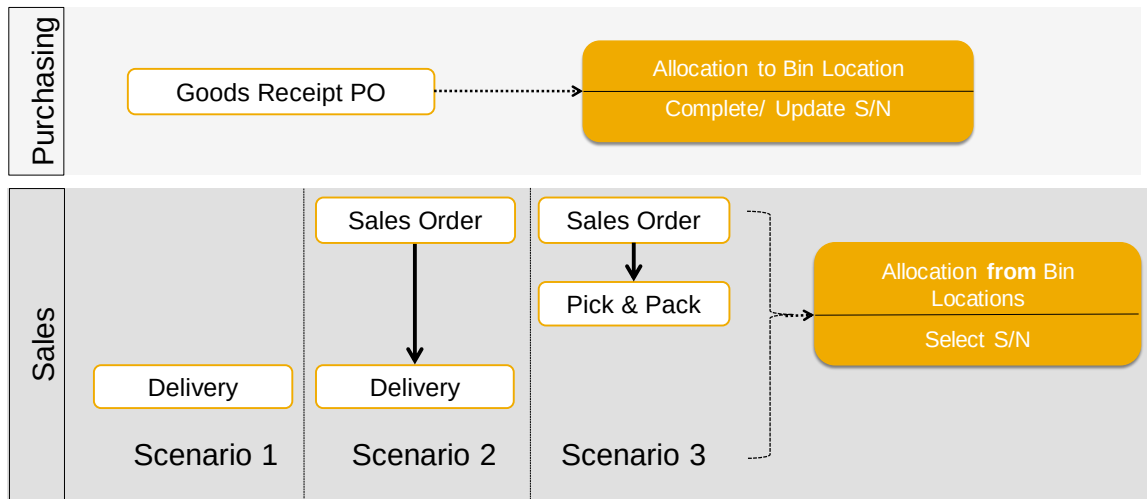


- You have finished configuring the bin location setup at OEC Computers.
- You have also guided OEC Computers employees through performing allocation processes and executing inventory reports for bin locations.
- Since OEC Computers purchases and sells many items managed by serial numbers and batches, you explain George, the warehouse manager, and his employees, about the allocation process of these items.

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Business Example

Business Process integrated with Bin Locations and Serial Numbers



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In this course we will look at some typical business scenarios in OEC Computers. These scenarios include purchasing, selling and picking items managed by serial numbers in a bin location managed warehouse that can be implemented at OEC Computers. We will start by adding a *Goods Receipt PO* for tablet devices which are also managed by serial numbers. In the *Goods Receipt PO* we will allocate the tablets to the desired bin locations and update their serial numbers.

After that, we will create three sales scenarios and examine different issue methods:

- In scenario 1, we will issue a *Delivery* document that is not based on another sales document.
- In scenario 2, we will see what happens when the *Delivery* document is based on a *Sales Order*.
- In scenario 3, we will see how the pick and pack process is influenced by the issue method.

Please note,

- To simplify the explanations in this course, the scenarios refer to serial numbers and not to batches throughout the training. Nevertheless, most explanations are relevant also to batches. We will however cover the key differences between serial and batch related processes.
- In addition the scenarios in this training refer to *Goods Receipt PO* and *Delivery* documents. However, the majority of the incoming and outgoing inventory processes are the same as with the *Goods Receipt PO* and the *Delivery* documents respectively.

Agenda

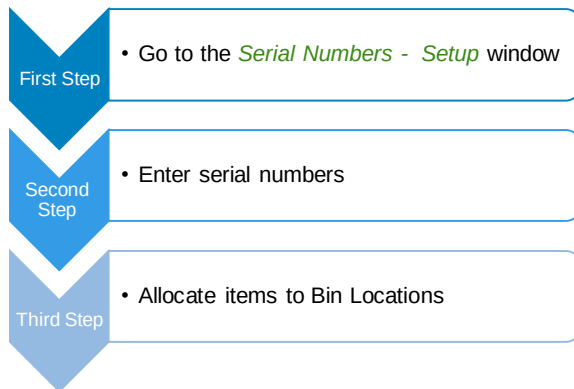
- Business example
- Purchasing process
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Let us examine a short purchasing process that demonstrates allocation of items managed by serial numbers in bin locations.

Serial Numbers in Incoming Bin Location Allocation

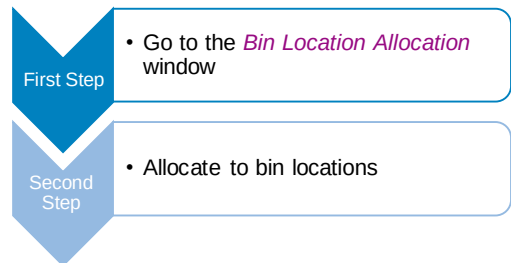
Method 1 – On every transaction

(Serial numbers are mandatory)



Method 2- Release only

(Serial numbers are not mandatory)



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This is a general schema of the working processes for incoming bin location allocation of serial numbered items.

As you are likely familiar, all items managed by serial numbers or batches have a management method assigned. The management method controls when serial numbers or batches are required.

The two methods are: *On every transaction* and *Release Only*.

As the graphic shows, the first method “On every transaction” will require you to assign a serial number when items are received into stock. When you create a goods receipt for the serial numbered item, the *Serial Number – Setup* window opens. There you assign the serial numbers and then the bin locations.

When you have a serial numbered item that is managed by *Release Only*, there is no requirement to enter a serial number when the item is received into stock. Therefore, when you create a *Goods Receipt PO*, you do not have to open the *Serial Number – Setup* window. Instead, you will use the *Bin Location Allocation* window, as you would with any other item not managed by serial numbers. If you do decide to add serial numbers when receiving an item managed by *Release Only* you can manually open the *Serial Number – Setup* window to enter serial numbers.

Note,

This is a general process only. The detailed working processes are affected when automatic allocations rules apply. In addition the *Automatic Serial Creation* definition in the *Item Master Data*, also affect working processes. Let us examine these working processes in the next slides.

Serial Numbers in Incoming Bin Location Allocation Method 1 - On Every transaction (1/2)

Item Master Data

Item No. Manual T00001
 Description Tablet 12" 32G
 Foreign Name
 Item Type Items
 Item Group Items
 UoM Group Manual
 Price List Base Price

General Purchasing Data Sales Data Inventory Data

☐ Do Not Apply Discount Groups

Manufacturer OEC Computers
 Additional Identifier
 Shipping Type Fedex EM

Serial and Batch Numbers
 Manage Item by Serial Numbers
 Management Method On Every Transaction

Issue Primarily By Bin Locations

Goods Receipt PO

Vendor V10000
 Name Acme Associates
 Contact Person Sarah Klerl
 Vendor Ref. No.
 BP Currency GBP

No. Primary 753
 Status Open
 Posting Date 08.09.
 Due Date 10.09.
 Document Date 08.09.

Contents Logistics Accounting Electronic Documents Attachments

Item/Service Type	Item	Quantity	Unit Price	Dis...	Tax ...	Total (LC)	Bin Location Allocat...	UoM Co...	Country of Origin	Summary Type	No Summary
1	T00001	5	GBP 250.00	0.00	11	GBP 1,250.00		Manual			
2				0.00	11						

In Method 1 - Bin location allocation is done only via the Serial Number - Setup window

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In the graphic we see an item which has the serial number management method *On Every Transaction*.

When this item is added to a *Goods Receipt PO*, we no longer have the option to open the *Bin Location Allocation – Issue/Receipt* window, like we would normally do with other items. As mentioned previously, we need to enter the allocation for the bin location from the *Serial Number – Setup* window, after entering the mandatory serial numbers.

However, after serial numbers are assigned and the allocation is made, we will be able to change the allocation by using the link in the *Bin Location Allocation* field to open the *Serial Number – Setup* window.

Let us see how the allocation process is done in the next slide.

Serial Numbers in Incoming Bin Location Allocation Method 1 - On Every transaction (2/2)

The screenshot shows the 'Serial Numbers - Setup' window. It has two main sections: 'Rows from Documents' and 'Created Serial Numbers'.

Rows from Documents:

#	Doc. ...	Item Nu...	Item Descri...	Whse ...	Whs...	Total Nec...	Total Cr...	Open Qty
1	PD 71	T00001	Tablet 12' 32G	01	Gen	5	2	3

Created Serial Numbers:

#	Mfr Serial No.	Serial Number	Lot Number	Bin Location	Expiration Date	Mfr Date	Admission Date	Mfr Warrant...
1	T42363678			01-SYSTEM-BIN-LOCAT			08.09.	
2	T42363679			01-SYSTEM-BIN-LOCAT			08.09.	
3							08.09.	

Annotations in the image:

- First Step:** Points to the 'Serial Number' column in the 'Created Serial Numbers' table.
- Second Step:** Points to the 'Bin Location' column in the 'Created Serial Numbers' table.

At the bottom, there is a 'Total Created' field with the value '2', and buttons for 'Update', 'Cancel', 'Reallocate Bin Locations', 'Automatic Allocation', and 'You Can Also'.

Method: On Every Transaction

1st Step ■ Enter a serial number

2nd Step ■ Bin location allocation

The image shows the two steps needed when receiving items with the management method *On Every Transaction*.

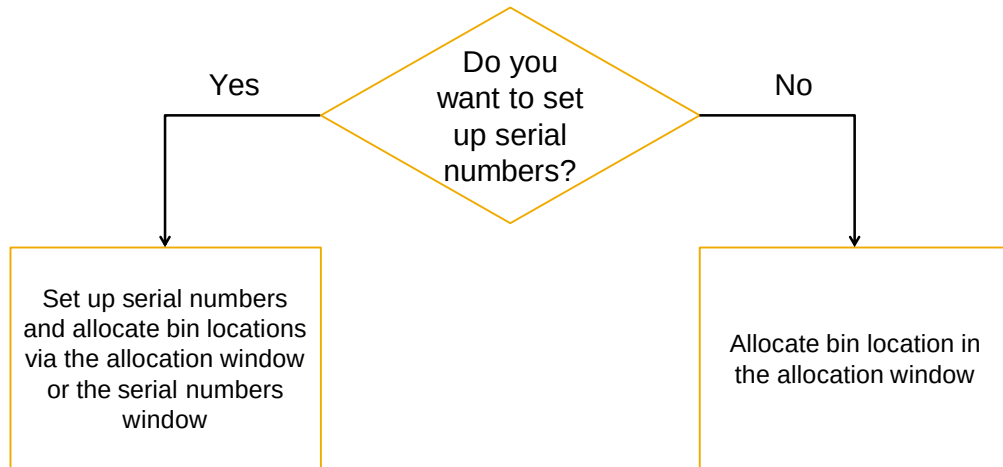
For every row of the document with an item requiring serial numbers, you enter the *Serial Number – Setup* window.

First, you enter the serial numbers. Once serial numbers are assigned, the system allocates a bin location automatically (assuming automatic allocation rules apply). You can also allocate a bin location manually.

Note,

- Documents may contain several rows. Some may be serials and some may not. When entering the *Serial Numbers – Setup* window, we see only the serial managed items.
- Further more, when choosing a non-bin location managed warehouse in the document row, none of the bin location related fields or buttons are visible in the window.

Serial Numbers in Incoming Bin Location Allocation Method 2 – Release Only



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For serial numbered items with the management method *Release Only*, you can assign numbers either in the usual way in the *Bin Location Allocation* window or when assigning the optional serial number through the *Serial Numbers – Setup* window.

If the *Serial Number Management* field in the *Item Master Data* record is set to *On Release Only* and the *Automatic Serial Number Creation On Receipt* box is checked, the system will automatically assign a bin location when adding the document, assuming automatic allocation rules apply. If no automatic receipt allocation rules apply then the incoming item is assigned to the system bin location.

Serial Number Setup Reallocate Bin Locations

The screenshot displays the 'Serial Numbers - Setup' window. It features a 'Rows from Documents' table at the top and a 'Created Serial Numbers' table below it. The 'Created Serial Numbers' table has columns for Mfr Serial No., Serial Number, Lot Number, Bin Location, Expiration Date, Mfr Date, Admission Date, and Mfr Warrant. Two rows are visible, both with '01-SYSTEM-BIN-LOCAT' as the bin location. A yellow arrow points from the 'Reallocate Bin Locations' button in the bottom right of the window to the 'Bin Location Reallocation for Serial Numbers' sub-window. This sub-window has a 'Find' field and a table with columns for Select, Mfr Serial No., Serial Number, Lot Number, Bin Location, and Expiration Date. Two rows are listed, with the first row selected. Below the table is a 'Reallocate to Bin Location' field with a dropdown menu showing '01-A001-S1-L1'. At the bottom of the sub-window are 'Update' and 'Cancel' buttons. A large yellow arrow labeled 'Reallocation' points from the sub-window back towards the main window.

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Sometimes a large quantity of an item is received and you would like to redistribute the quantity into multiple bin locations.

In the *Serial Numbers – Setup* window there is a way to distribute items from the *Created Serial Numbers* grid into bin locations.

Choose the *Reallocate Bin Locations* button to open the *Bin Location Reallocation for Serial Numbers* window. The list of items is then copied to this window.

To trigger the reallocation, just select the serial numbers to reallocate and choose the target bin in the *Reallocate to Bin Location* field. After update, the new bin locations appear in the *Serial Numbers – Setup* window.

OEC Computers purchases large quantities of a tablets. Each tablet has a unique serial number and the quantity is too large for just one bin location. The warehouse employee opens the *Serial Numbers – Setup* window and updates the serial numbers for the whole quantity. When he finishes, he chooses the *Reallocate Bin Locations* button to open the *Bin Location Reallocation for Serial Numbers* window. Here he selects about 20 units and allocates them to a bin location. Then he selects another 20 and allocates them to another bin. The employee repeats this action until everything is allocated. This is an easy way to allocate large quantities of the same item rather than choosing the bin location in every serial row in the *Created Serial Numbers* grid.

Note, reallocation is possible both for items which were already allocated and for those which were not allocated yet.

Serial Number Setup

Automatic Serial Number Creation

Serial Numbers - Setup

#	Doc	Item No.	Item Description	Whse	Total New	Total Cr.	Open Qty
1	PD 71	T00001	Tablet 12' 32G	01	5	2	3

Created Serial Numbers

#	Mfr Serial No.	Serial Number	Lot Number	Bin Location	Expiration Date	Mfr Date	Admission Date	Mfr Warranty
1	142363678			01-AD01-S1-L1	08.09.			
2	142363679			01-SYSTEM-BIN-LOCAT	08.09.			
3								

Total Created: 2

Buttons: Update, Cancel, Breakdown Bin Locations, Automatic Allocation, You Can Also

Automatic Serial Numbers Creation

Item Number: T00001
 Item Description: Tablet 12' 32G
 Warehouse: 01
 Bin Location:
 Quantity: 3
 Mfr Serial No.:
 Serial Number:
 Lot Number:
 System No.:
 Admission Date: 08.09.
 Manufacturing Date:
 Expiration Date:
 Mfr Warranty Start:
 Mfr Warranty End:
 Location:
 Details:
 Buttons: Create, Cancel

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As always with receiving serialized items, you have the option to have the system generate serial numbers for incoming items so that you do not need to enter the numbers manually. To do this, choose the *Automatic Creation* button to open the *Automatic Serial Numbers Creation* window.

The *Bin Location* field allows selecting a bin location from a list filtered by the *Warehouse* field.

Entering data in this field is optional. If it is populated then the same bin location code will be applied to all serial number rows created by this window.

The *Bin Location* field will be updated automatically with the *Default* bin location if one exists.

Complete Serial Numbers

From the incoming inventory document

Document No. 756
 Row No. 1 of 1 Rows
 Warehouse Code 01
 Item No. 100002
 Quantity 2
 Total Allocated 2
 Remaining 0

#	Bin Location	Allocated
1	01-SYSTEM-BIN-L1	1
2	01-A001-S1-L1	1
3		

Buttons: Update, Cancel, Automatic Allocation, Clear Allocations

Inventory → Item management → Item serial Numbers → Serial Number management → Complete Mode

Rows from Documents

#	Doc No.	Item Number	Item Description	Warehouse Code	Warehouse	Total Recd.	Total Created	Open Qty
1	PO 748	1001	Tablet 12" 64GB	01	General	100	2	
2	PO 756	100002	Tablet 12" 64GB	01	General	2		

Created Serial Numbers

#	Item Serial No.	Serial Number	Lot Number	Bin Loc.	Expiration Date	Mfg. Date	Admission Date	M.
1		2525100					08.09	

List of Bin Locations

#	Bin Location	Alternative Sort Code
1	01-A001-S1-L1	
2	01-SYSTEM-BIN-LOCATION	

Available bins to choose from

Buttons: Choose, Cancel, New

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Some items that OEC purchases, such as tablet computers, are received into the warehouse without serial numbers. However, the company likes to assign serial numbers to them shortly after, for tracking purposes.

In those cases, the items are set up with the management method *On Release Only* and received without serial numbers into bin locations.

Later, George assigns serial numbers to these items using the *Serial Number Management – Complete* window.

This window is accessed from the *Serial Numbers Management* window.

We can see an example of this in the image.

A *Goods Receipt PO* number 756 was created for 2 tablet units.

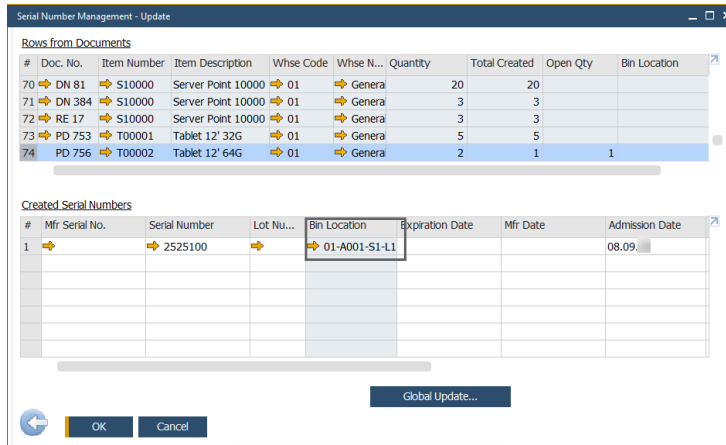
Those 2 tablets were allocated to 2 different bins.

The next step is to complete the serial numbers for the items and relate each serial to the relevant bin.

In the *Serial Number Management – Complete* window, in the *Created Serial Numbers* grid, we first enter the serial number and then we can choose the relevant bin location from the list of bins entered for the item in the *Goods receipt PO*.

When only one bin is chosen in the *Goods Receipt PO* for the whole quantity, then this bin is updated automatically in the *Bin Location* column of the *Created Serial Numbers* grid.

Update Serial Numbers



#	Doc. No.	Item Number	Item Description	Whse Code	Whse N...	Quantity	Total Created	Open Qty	Bin Location
70	DN 81	S10000	Server Point 10000	01	Genera	20	20		
71	DN 384	S10000	Server Point 10000	01	Genera	3	3		
72	RE 17	S10000	Server Point 10000	01	Genera	3	3		
73	PD 753	T00001	Tablet 12' 32G	01	Genera	5	5		
74	PD 756	T00002	Tablet 12' 64G	01	Genera	2	1	1	

#	Mfr Serial No.	Serial Number	Lot Nu...	Bin Location	Expiration Date	Mfr Date	Admission Date
1		2525100		01-A001-S1-L1			08.09.

Global Update...

OK Cancel



Changes or updates to existing serial numbers created can be done in the Update mode of the *Serial Number Management* window.

The bin location code however can not be changed.

The read only *Bin Location* column allows you to view in which bin location each serial is located.

Agenda

- Business example
- Purchasing process
- Sales process
- Differences in Batches

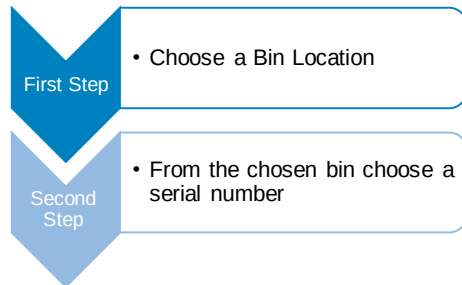
Let us see the changes made in the sales processes.

Priority of Number or Bin Location Selection

Introducing the Issuing Methods

Method 1- Issue Primarily By:

Bin Locations



Method 2 - Issue Primarily By:

Serial and Batch Numbers



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In OEC Computers, some items are picked according to their serial numbers and not according to their bin location. For example, OEC Computers resells custom-made servers according to special configuration ordered by the customer. All these servers have the same item code and every unique server is identified by its serial number. When the server is delivered to the customer, the server is picked according to its serial number.

In SAP Business One, we can identify the picking method needed for these items by indicating the appropriate issuing method on the item master data record:

- Method 1 – *Issue Primarily By Bin Locations.*

In this method you first choose the bin location to allocate from and then choose a specific serial item stored within the chosen bin.

- Method 2 - *Issue Primarily By Serial and Batch Numbers.*

In this method you choose the serial number and then consequently the bin location the item is stored in.

In OEC Computers, almost every item issue method is set to method 1 – *Issue Primarily by Bin Locations*, due to the following reasons:

- First, in most cases when selling products, it does not matter to the customer nor to OEC Computers which exact serial number to issue.
- Second, this method allows easy and fast picking because there is no need to look for a specific serial number.
- In addition, this method enables maximum control of the storage arrangement in the different bin locations.

Note that these two methods also affect scenarios where serial number selection and bin location allocation are not performed in the same document. For example, issuing a *Sales Order* with serials and then copying it to a *Delivery* where the bin location allocation is done. We will explore this scenario and another scenario involving the *Pick and Pack* procedure later on in this training.

Priority of Number or Bin Location Selection Definitions

The image shows two SAP configuration screens. The left screen is 'General Settings' under the 'Inventory' tab, specifically the 'Serial Numbers and Batches' section. It shows the 'Issue Primarily By' field set to 'Serial and Batch Numbers'. The right screen is 'Item Master Data' for item 'T00001'. It shows the 'Manage Item by' field set to 'Serial Numbers' and the 'Issue Primarily By' field set to 'Bin Locations'. Both fields are highlighted with orange boxes.

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Let us see how we define the issue method in the system.
the issue method is defined in the *Item Master Data* and can be changed manually. The value in this field is defaulted from the **General Settings → Inventory → Items → Issue Primarily By field**.

Note, the settings of the field in *Item Master Data* cannot be changed if open sales orders with serial or batch allocations exist.

In the next slides we will examine 3 scenarios:

- Scenario 1 - a delivery not based on a sales order
- Scenario 2 – a delivery based on a sales order
- Scenario 3 – *Pick and Pack* procedure

Scenario 1 – Issuing a Delivery Not Based on a Sale Order

The screenshot shows a SAP Delivery document for Customer C23900 (Parameter Technology). The document is titled 'Delivery' and has a primary key of 1362. The status is 'Open' and the posting date is 08.09. The delivery date is also 08.09. The currency is GBP.

Two items are listed in the table:

Item No.	Quantity	Unit Price	Disco...	Tax ...	Total (LC)	Whse	Bin Location Allo...	UoM Code	COGS Departments
1	1	0.00	01	0.00	01	01	1	Manual	
2	1	0.00	01	0.00	01	01		Manual	

Below the table, the 'Sales Employee' is Brad Thompson. The 'Total Before Discount' is 0.00, and the 'Total' is 0.00. The 'Issue Primarily By' dropdown is set to 'Serial and Batch Numbers' for Item 1 and 'Bin Locations' for Item 2.

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Let us go back to OEC Computers. In the first scenario we issue a delivery not based on a sales order.

OEC Computers sell their customer, Parameter Technology, two items.

The first is a custom-made server that is issued primarily by serial numbers.

The second item is a tablet item which is also managed by serial numbers but with an issue method primarily by bin locations.

Let us see the differences in the allocation processes of these items.

Look at the server item row. The *Bin Location* column is blocked and the only way to allocate the item from a bin location is by entering the *Serial Number Selection* window through the *Quantity* field.

In contrast, the tablet item can be allocated from a bin location also by choosing the link arrow icon in the *Bin Location Allocation* column. It can also be allocated automatically assuming automatic allocation rules apply. In the image we see the tablet item was already allocated automatically.

Note that when working with issue method *Primarily By Serials and Batches*, you need to first choose the serial numbers and then the associated bins will be allocated as well.

In the next slide we will see how the allocation is made for each item.

Scenario 1 – Issuing a Delivery Not Based on a Sale Order

Serial Numbers Selection Window – Issue Primarily By SN

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The *Serial Number Selection* window supports allocating bin locations for both issue methods.

In the *Available Serial Number* grid, we can display the different sub level codes, bin location attributes and alternative sort code.

The primary purpose of these fields is to assist the *Auto Select* functionality. The *Auto Select* simply selects serial numbers for the quantity required according to the current sort order of the *Available Serial Numbers* grid. The serial number selection can be managed by bin location sort sequences or attributes, even for an item which is set to issue primarily by serial number.

The different functionalities available in the window behave differently according to the issue method of the item chosen in the *Rows from Documents* grid.

In the image the server item is chosen which is issued primarily by serial numbers. Look at the available serial numbers for the server. The *Serial Number* column is displayed followed by the *Bin Location* Column.

There are *Filter* and *Find* functions in the *Available Serial Numbers* header that allow selecting the relevant field to be filtered by and filtering the list displayed accordingly. The *Filter By* field supports the selection of the different serial number properties: Serial Number, Lot Number, Manufacturer Serial Number and System Number. When selecting the serial number the associated bin location is selected automatically.

Scenario 1 – Issuing a Non-Based Delivery Document

Serial Numbers Selection Window – Issue Primarily By Bin Locations

The screenshot displays the 'Serial Number Selection' window. At the top, the 'Rows from Documents' table shows two rows of items: 'Tablet 12' 32G' and 'Tablet 12' 64G'. The 'Issue Primarily By' dropdown is set to 'Bin Locations'. Below this, the 'Available Serial Numbers' section shows a table with columns for 'Serial Number', 'Bin Location', and 'Allocated'. The 'Bin Location' column is highlighted. The 'Selected Serial Numbers' section shows a table with columns for 'Bin Location', 'Floor', 'Area', 'Row', and 'Shelf'. The 'Auto Select' button is visible at the bottom right, with a dropdown menu showing options: 'Single Choice', 'Bin Location Code Order', 'Alternative Sort Code Order', 'Descending Quantity', 'Ascending Quantity', 'FIFO', and 'LIFO'.

#	Item No.	Item Description	Whse Code	Quantity	Total Selected	Open Qty	Direction
1	T00001	Tablet 12' 32G	01	1	1		Out
2	T00002	Tablet 12' 64G	01	1		1	Out

#	Serial Number	Bin Location	Allocated
1	2525100	01-A001-S1-L1	No

#	Bin Location	Floor	Area	Row	Shelf
---	--------------	-------	------	-----	-------

Available Quantity: 1
 Display Available SNs: All
 OK Cancel Auto Select

Auto. Selection
 Single Choice
 Bin Location Code Order
 Alternative Sort Code Order
 Descending Quantity
 Ascending Quantity
 FIFO
 LIFO

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Let us examine what happens to the *Serial Number Selection* window when choosing an item with the issue method *Primarily by Bin Locations*.

In our example we have chosen the tablet item.

Here, the *Filter* fields allow filtering by the bin location codes or warehouse sublevels.

The *Bin Location* Column is displayed first followed by the serial Number column.

Instead of the *Auto Select* button there is a *Selection Method* button. This button provides the regular bin location selection options in addition to the *Auto Selection* option.

The Auto Select option selects bin locations according to their order of appearance in the Available Serial Numbers grid.

Scenario 2 – Delivery Based on a Sales Order

	Issue Primarily by Bin Locations	Issue Primarily by SN and Batches
Serial selection in a Sales Order	Not possible	Possible
Allocation of serial numbers and bin locations in a delivery based on a sales order	Regular allocation according to the Issue Primarily by Bin Locations rules	If serials <u>were not</u> selected then select serials and the associated bin location If serials <u>were</u> selected in the Sales Order then the Bin Location is selected automatically in the document.

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In the second scenario, a delivery is issued based on a sales order.

In most cases, the process for allocating bin locations in a delivery will remain the same whether or not the delivery is based on the sales order. However there can be circumstances that affect allocation in the delivery.

First let us look at issuing items primarily by bin locations. It is not possible to select serial numbers in a sales order for items set to this method. Therefore the allocation process in a Delivery document based on the sales order remains the same as described in the former slides.

For items that are issued primarily by serial numbers, it is possible to select the serial numbers in the sales order.

If serial numbers are not selected in the sales order, then the allocation process for the delivery remains the same as described in the former slides.

However, if serial numbers were selected in the sales order, then the bin locations associated with the selected serial numbers are automatically allocated in the document.

Scenario 3 - Serial Numbers in *Pick and Pack*

Function	Primarily By Serial and Batch Numbers	Primarily By Bin Locations
Select Serials in the Pick and Pack Manager	Possible	Not possible
Generation wizard	Generates pick list only for orders lines already containing serials	Generates pick list for all order lines
Generation wizard filtering	Filter bin locations by selected serial numbers	Subject to the regular wizard filtering
Released Qty field in the Released Drawer	Pick list cannot be edited or processed unless there is a full allocation of both SN and bin locations	Regular pick list behavior, without serial allocation
Creating a Delivery based on a picked Sales Order	Serial numbers should be selected first for the Sales Order	Serial numbers are selected when creating the Delivery
Bin location allocation changes in the pick list before the items are picked	Possible to open the <u>Select Serial Number</u> window and edit the selection of both serial numbers and bin locations	Possible to open the <u>Bin Location Allocation</u> window and edit the bin location allocation

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The third scenario refers to the *Pick and Pack* process.

This process changes according to the “Issue primarily by” method of each item involved in the process.

The table displayed here lists the main differences between each issuing method in the pick and pack processes.

For items issued primarily by serial numbers, all pick and pack functions can be performed only if serial numbers have been selected. For example, generating pick lists and creating a delivery is only possible when the serial numbers have already been specified in the sales order. If serials were not selected and the *Release to Pick List button* was chosen, then the system opens the *Serial Number Selection* window and thus enables selecting serials during the *Pick and Pack* procedure.

For items issued primarily by bin locations, most pick and pack functions can be done in the regular way. It is not possible, however, to select serial numbers before a delivery document is created.

Note that any differences that refer to creating deliveries are relevant also for creating invoices.

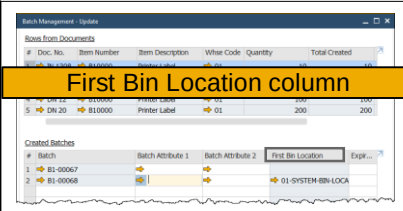
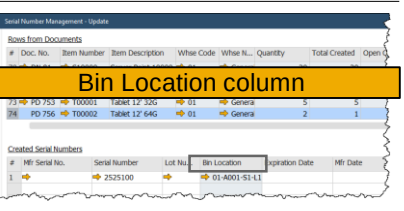
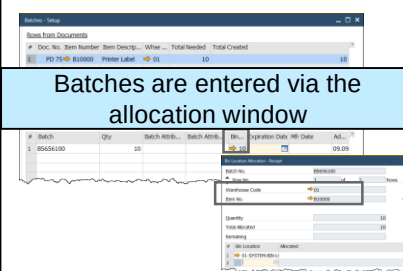
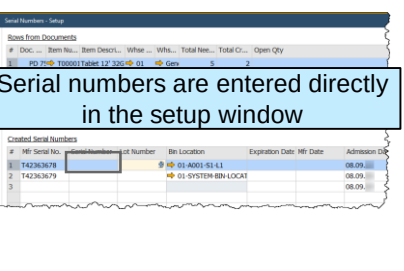
Agenda

- Business example
- Purchasing process
- Sales process
- Differences in Batches

Finally, we will go over the main differences in processes involving batch numbers instead of serial numbers.

In addition we will examine a scenario for working with Batches using the FIFO allocation method.

Functionality Differences for Batch Related Items (1/2)

	Batches	Serials
Difference No. 1	 First Bin Location column	 Bin Location column
Difference No. 2	 Batches are entered via the allocation window	 Serial numbers are entered directly in the setup window

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The purchasing and sales processes for batch-managed items in a bin location managed warehouse is similar to the processes of serial number managed items. In this section, we will review some of the differences between processes based on batch items and processes based on serial items.

Look at difference number one. This is one of the more obvious differences. For batch managed items, the *First Bin Location* column is used instead of the *Bin Location* column. Since one batch number represents a group of items and not just a single item as in serial numbers. When choosing the link arrow of the *First Bin Location*, the *Bin Location Content List* opens filtered by the chosen batch.

Look at difference number two. Since one batch can be stored in multiple bin locations, in the *Batch - Setup* window as well as in the *Batch Complete* and *Update* windows, the batch selection is done via the *Bin Location Allocation – Receipt* window. In the *Batch Setup* window, the link arrow in the *Bin Location* field opens the *Bin Location Allocation – Receipt* window.

Functionality differences for Batch related items (2/2)

Batches

No Reallocation option

Serials

Reallocation is possible

Difference No. 3

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The last difference discussed in this training is the reallocation option in the setup window. The reallocation option we saw earlier in this training in the *Serial Numbers – Setup* window, does not exist in the *Batches - Setup* window.

The main reason for this behavior is that unlike serial numbers, where each item purchased has its own serial number, a few batch numbers are usually associated with large quantity of items.

For example, in OEC Computers, the Printer Label item is batch managed. The printer labels are purchased in dozens but each 10 units batch has one batch number. Therefore, in the example shown here, there are only five rows in the *Created Batches* grid and the allocation function is redundant.

First Bin Location in Batch Number Selection

Batch Number Selection

Rows from Documents

#	Item No.	Item Description	Whse C...	Quantity	Total Needed	Total Selected	Total Batches	Direction
1	B10000	Printer Label	01	10	10	10	2	Out

First Bin Location

#	Bin Location	Batch	Availabl...
1	01-SYSTEM-BIN-LOC	B1-00069	5
2	01-SYSTEM-BIN-LOC	B1-00070	8.333
3	01-SYSTEM-BIN-LOC	B1-00071	8.333
4	01-SYSTEM-BIN-LOC	B1-00072	8.333
			63.331

Selected Batches

#	Bin Location	Batch
1	01-SYSTEM-BIN-LOC	B1-00068
2	01-SYSTEM-BIN-LOC	B1-00069

Display Available Batches: All

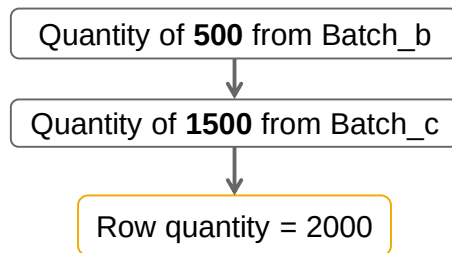
OK Cancel Selection Method

- First Bin Location column is available also for items managed by batch number first.
- Enable choosing items by both batch number and bin location

In some cases, when working with batch managed items, there is a need to pick items by both issuing methods: bin location first and batch number first. Therefore a *First Bin Location* column exists also in the *Available Batches* table in the *Batch Number Selection* window – even for items which are primarily selected by batch and serial numbers. The *First Bin Location* column is also available in the *Sales Order*. Please note that this column is not visible by default.

Using the FIFO Automatic Allocation Method with Batches

Bin#	Batch#	Stored QTY of Item A	Entrance date of the batch
Bin_1	Batch_a	1000	1.7.
	Batch_b	500	1.5.
Bin_2	Batch_c	2000	1.6.



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In the **Allocation processes in Bin Locations** course, we were introduced to the FIFO-LIFO automatic allocation methods for outbound transactions.

Companies that work mainly with items managed with batch numbers, can find these methods very useful.

If the company's policy is to issue the oldest items first then a good solution will be to use the FIFO allocation method.

In the FIFO method the items are allocated according to the entry date of the item per bin location per batch or serial number.

Let us examine the scenario demonstrated here.

Item A is stored in two bin locations.

Bin_1 contains two batches: batch_a and batch_b.

Bin_2 contains batch_c.

Let us say we need to issue a quantity of 2000 units of item A and we are using the FIFO method.

The system looks for the batch with the earliest entry date. This will be Batch_b.

Next, the system should allocate another 1,500 units and it looks for the batch with the next earliest date in sequence which is Batch_c.

Even though Batch_a is located in the same bin location as Batch_b the system knows how to trace the specific batch.

Summary



Here are some key points for working with serial numbers and batches in a bin locations managed warehouse:

- With incoming allocation, when the management method of the serial or batch number is set to *On Every Transaction* – then bin locations can be allocated only in the *Serial Numbers – Setup* window.
- In order to shorten a receipt process of serial or batch numbered items, you can first allocate the entire quantity to one bin location and then reallocate to different bin locations using the *Bin Location Reallocation* window.
- In the *Item Master data*, define the *Issue Primarily by* method: bin locations or serial and batch numbers. Choose the second method if you need to choose a specific serial number or batch.
- When the issuing method is set to *Issue Primarily by Bin Locations*, you cannot assign Serial or batch numbers in a sales order, only later on in a delivery or invoice.

Here are some key points you can take away from this training:

- With incoming allocation, when the management method of the serial or batch number is set to *On Every Transaction* – then bin locations can be allocated only in the *Serial Numbers – Setup* window.
- In order to shorten a receipt process of serial or batch numbered items, you can first allocate the entire quantity to one bin location and then reallocate to different bin locations using the *Bin Location Reallocation* window.
- In the *Item Master data*, define the *Issue Primarily by* method: bin locations or serial and batch numbers. Choose the second method if you need to choose a specific serial number or batch.
- When the issuing method is set to *Issue Primarily by Bin Locations*, you cannot assign Serial or batch numbers in a sales order, only later on in a delivery or invoice.

Summary



- The same rule applies in the Pick and Pack process – selecting serial or batch numbers are available only when the issuing method is set to primarily by serial numbers and batches.
- The issuing method also affect the pick list generation and the working procedures in the *Pick Pack and Production* wizard.
- When working with batch numbers, since one batch contains several items, instead of the *Bin Locations* filed, the *First Bin Location* displays where the first item of the batch was allocated.
- When choosing the link arrow of the *First Bin Location in a document*, the *Bin Location Content List* opens filtered by the chosen batch.
- In the FIFO issuing method, the items are allocated according to the entry date of the item per bin location per batch or serial number and not just per bin location. This method assists in releasing older batches first.

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- The issuing method also affect the pick list generation and the working procedures in the *Pick Pack and Production* wizard.
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- In the FIFO issuing method, the items are allocated according to the entry date of the item per bin location per batch or serial number and not just per bin location. This method assists in releasing older batches first.

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